

terminals for artillery coordination, supporting combat communication, and controlling the UAVs. There have been reports of the terminals aiding the Ukrainian armed forces in making precision strikes against Russian tanks and positions, and to monitor and coordinate UAVs to fire anti-tank weapons with targeted precision. Subsequently, when the Russian forces managed to jam the terminals, Starlink quickly updated the software to restore the services.

The Chinese concerns

Military observers all over the world, particularly in China, are increasingly getting concerned about the US using the satellite internet for dominance in space. They have contended that the orbital space should not become the exclusive privilege of the US and no single country

should have complete dominance in space.

China has complained to the UN Space Committee that the US is using a civil cloak to provide seamless reconnaissance and surveillance services for their armed forces. China has expressed the fear that it is a matter of deep concern that the Starlink is increasing the satellites from 12,000 to 42,000.

The system supposedly developed for civilian use with the US funding and infrastructural support could be reconfigured quickly for defence use. As an example, China has said that their Tiangong space station had two near-misses with Starlink satellites. As a countermeasure, China is deliberating developing capabilities to disable and may be even destroy Starlink internet satellites.

Risks and challenges ahead

Geostationary satellites can cover huge areas with a single satellite, whereas the LEO satellites need to be deployed in very large numbers to provide the coverage for uninterrupted continuous transmission. With thousands of satellites being launched into the lower orbit of space, there is growing concern among astronomers, scientists, and space agencies.

The issues flagged are that they ruin images or spectroscopic measurements as they streak through the telescope's field of view, making it difficult to observe the



GSAT-7B being built by ISRO will provide the Indian Army with high-speed satellite internet (Source: <https://dhyeyaias.com>)

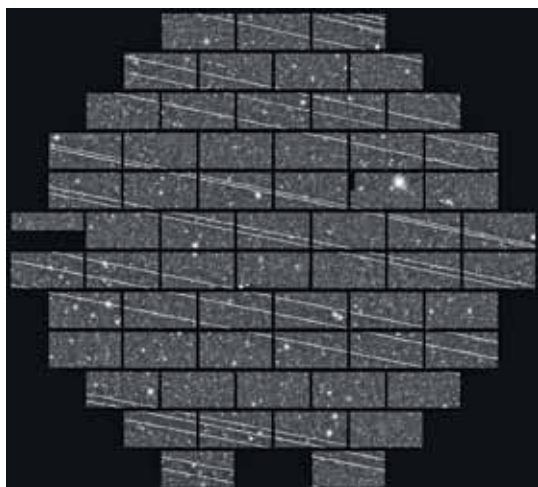
other space objects and detect their signals. They create a significant number of space debris, increasing the risk of collisions. They increase light pollution as the light reflected from the satellites could interfere with and be mistaken for the light coming from heavenly bodies. And the optical and radio waves' brightness severely affects the scientific observations.

For example, in 2019, the Blanco telescope, Cerro Tololo Inter-American Observatory (CTIO), reported strong signal loss and appearance of 19 white lines on the photo-shot of Dark Energy Camera (DECam)—a state-of-the-art camera for observing galaxies and recording motion—due to transit of a Starlink satellite train. To obviate their fears, Starlink has taken several steps, such as experimental coating trial, but the satellites were found to be still too bright. Thereafter, SpaceX started providing sunshades to reduce the brightness by about 30%.

Starlink is now providing satellite tracking data to astronomers and on-demand orientation adjustment for effective coordination. In addition, The International Astronomical Union (IAU) has established a centre for developing software tools and the relevant technologies to deal with the adverse effects of satellite constel-



SpaceX deployed 10,000 satellite terminals to restore destroyed internet connections in Ukraine (Source: <https://mil.in.ua>)



Strong signal loss and appearance of white lines due to transit of Starlink satellites on an image taken at the Cerro Tololo Inter-American Observatory (CTIO) (Source: <https://physics.aps.org>)